

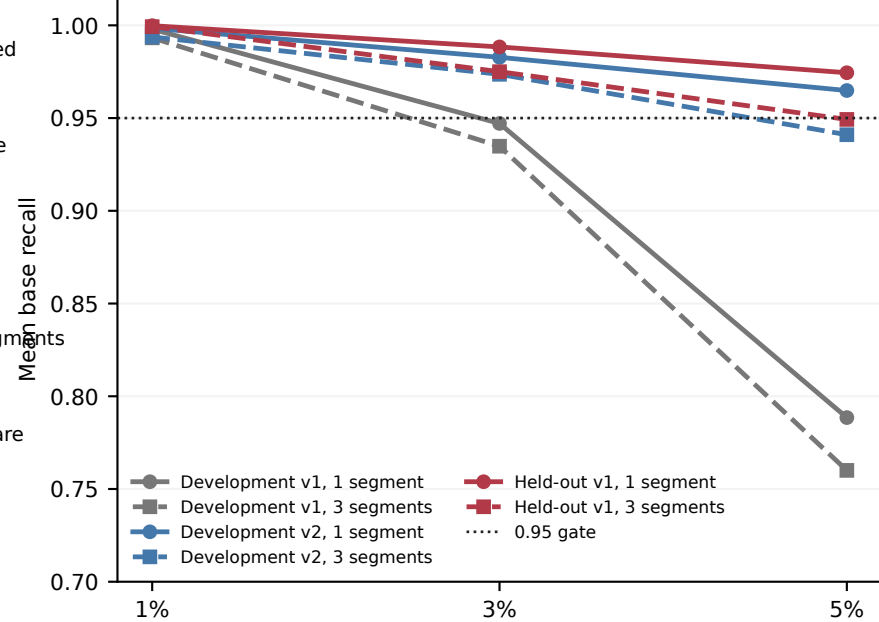
Divergence-aware anchor bridging passes predeclared localization gates

A Predeclared validation design

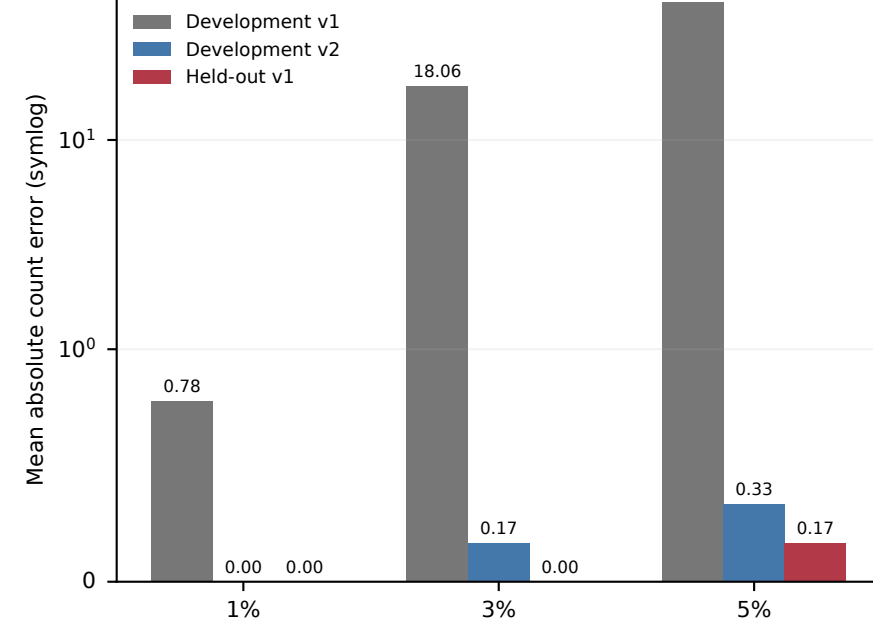
Failed development	3	seeds 5301-5303, v1 retained
Selected development	3	same seeds, bounded bridge
Held-out genomes	3	fresh seeds 5501-5503
Array scenarios	6	1/3/5% divergence x 1/3 segments
Held-out commands	1,062	localize, quantify and compare

Known catalogue; substitutions only; no biological truth

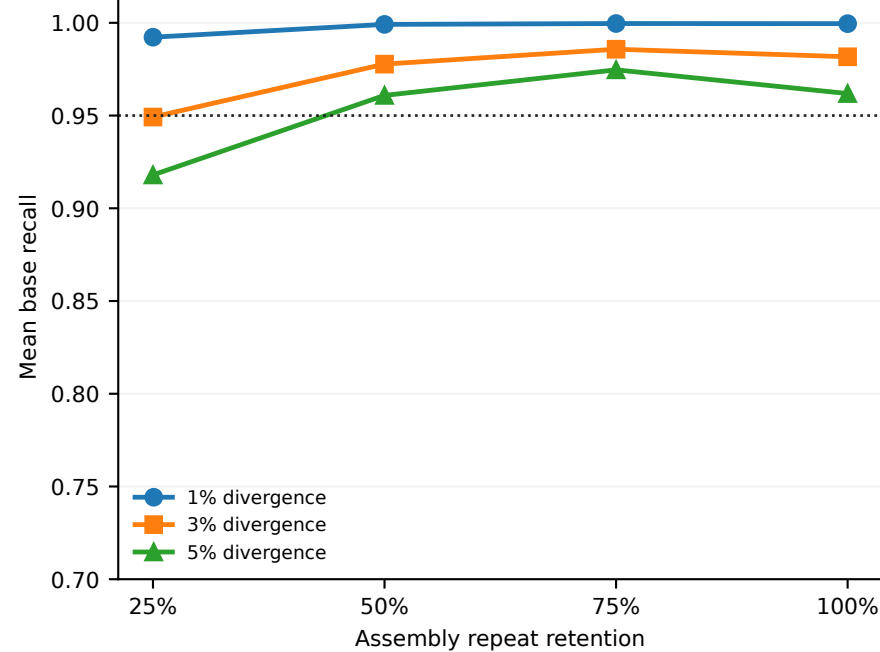
B Full-assembly localization recall



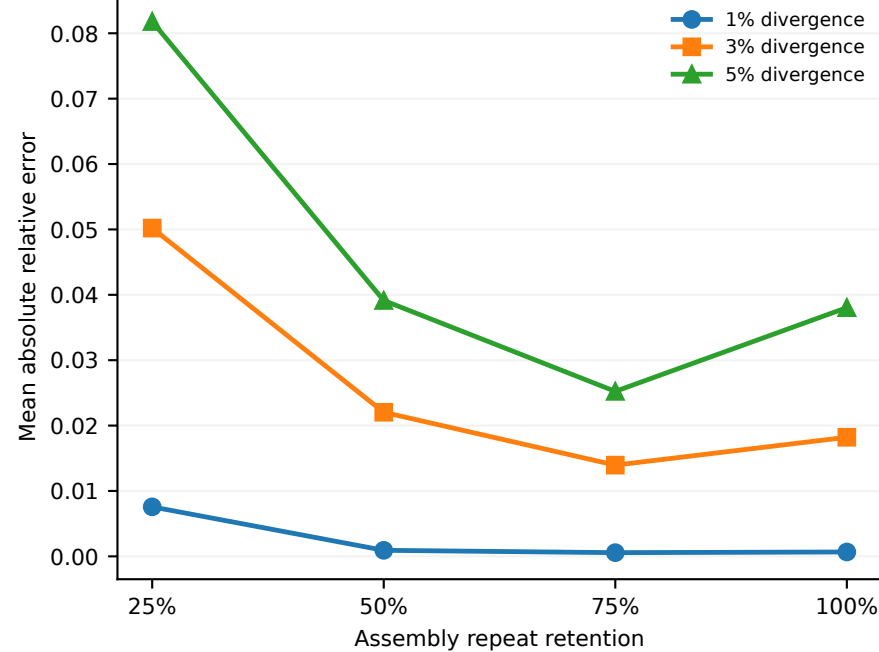
C Predicted array fragmentation



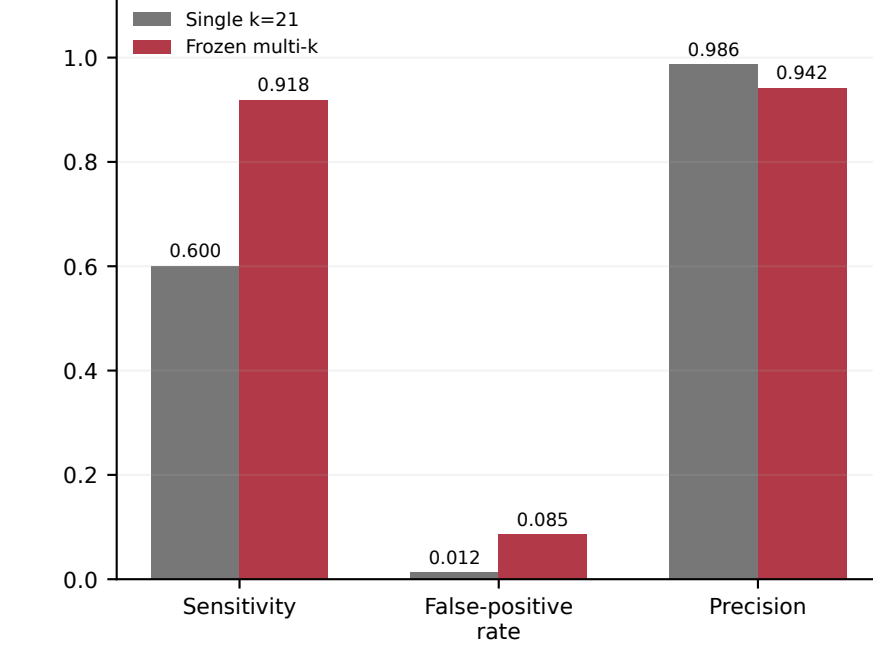
D Held-out recall across collapse



E Held-out localized repeat-bp error



F Assembly/read classification



Fresh seeds validate localization; the frozen multi-k classifier gains sensitivity while increasing false positives.